USER MANUAL

IoT-Based Smart Home Automation & Safety System

HiveLink

HOME AUTOMATION

Hive Link

Intelligent Home Automation & Safety System

"Making Homes Safer, Smarter, and More Efficient."

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Welcome to HiveLink

Dear HiveLink User,

Welcome to a new era of intelligent living. HiveLink is not just a product - it is your home's smartest companion, purpose-built to protect, automate, and empower every corner of your living space.

Designed with cutting-edge IoT technology, HiveLink seamlessly integrates real-time sensor monitoring, automated controls, remote access, and intelligent safety alerts - all working in harmony to give you peace of mind, whether you are at home or miles away.

From your kitchen's temperature to your garage's parking precision, from gas leak alerts to keyless RFID entry - HiveLink handles it all, silently and efficiently, so you can focus on what matters most.

This User Manual will guide you through everything you need to know about setting up, operating, and getting the most out of your HiveLink system. We recommend reading through it carefully to unlock the full potential of your smart home.

Thank you for choosing HiveLink - where technology meets home.

The HiveLink Team

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Why Choose HiveLink?

HiveLink is engineered around five core pillars that define the modern smart home experience:

SFT

Safety First
HiveLink's gas leak detection and real-time alerts ensure your family is protected 24/7 from kitchen hazards and fire risks.

AUTO

Intelligent Automation
Smart lighting, ventilation, and parking assistance work automatically — reducing manual effort and human error.

SEC

Enhanced Security
RFID-based door locking eliminates traditional key vulnerabilities, ensuring only authorized individuals gain access.

NRG

Energy Efficiency
LDR-based outdoor lighting activates only when needed, and smart ventilation runs only when thresholds are breached — saving electricity intelligently.

CON

Ultimate Convenience
The Blynk IoT mobile app gives you real-time monitoring and remote control from anywhere in the world, right from your smartphone.

Product Features

HiveLink delivers a comprehensive suite of seven intelligent features, powered by two microcontrollers working in perfect synchronization.

ESP32-POWERED FEATURES

[TMP]

Smart Kitchen Monitoring

Continuously monitors kitchen temperature and humidity via DHT11 sensor. Automatically activates a 12V DC ventilation fan when temperature exceeds 30 degrees C or humidity exceeds 60%.

[GAS]

Gas Leak Detection System

MQ-5 gas sensor detects LPG and combustible gases in real time. Triggers an audible buzzer and sends instant ALERT notifications to the LCD display and Blynk mobile app when gas levels exceed the safe threshold.

[RFC]

RFID Smart Door Lock

Contactless RFID card and tag authentication replaces traditional keys. Only pre-authorized RFID credentials unlock the door, keeping unauthorized entries blocked.

[LCD]

Real-Time LCD Display

A live dashboard display showing temperature, humidity, gas value, gas status (SAFE / ALERT), and fan status (ON / OFF) - all updated in real time.

[APP]

Blynk IoT Mobile App

Full Wi-Fi-connected remote monitoring. View all sensor values, control lights remotely, and receive gas status alerts — all from the Blynk smartphone application.

ARDUINO UNO-POWERED FEATURES

[LDR]

Smart Outdoor Lighting

LDR sensor automatically switches fence and rooftop lights ON when ambient light falls below the threshold, and OFF in daylight — fully automatic, zero manual effort.

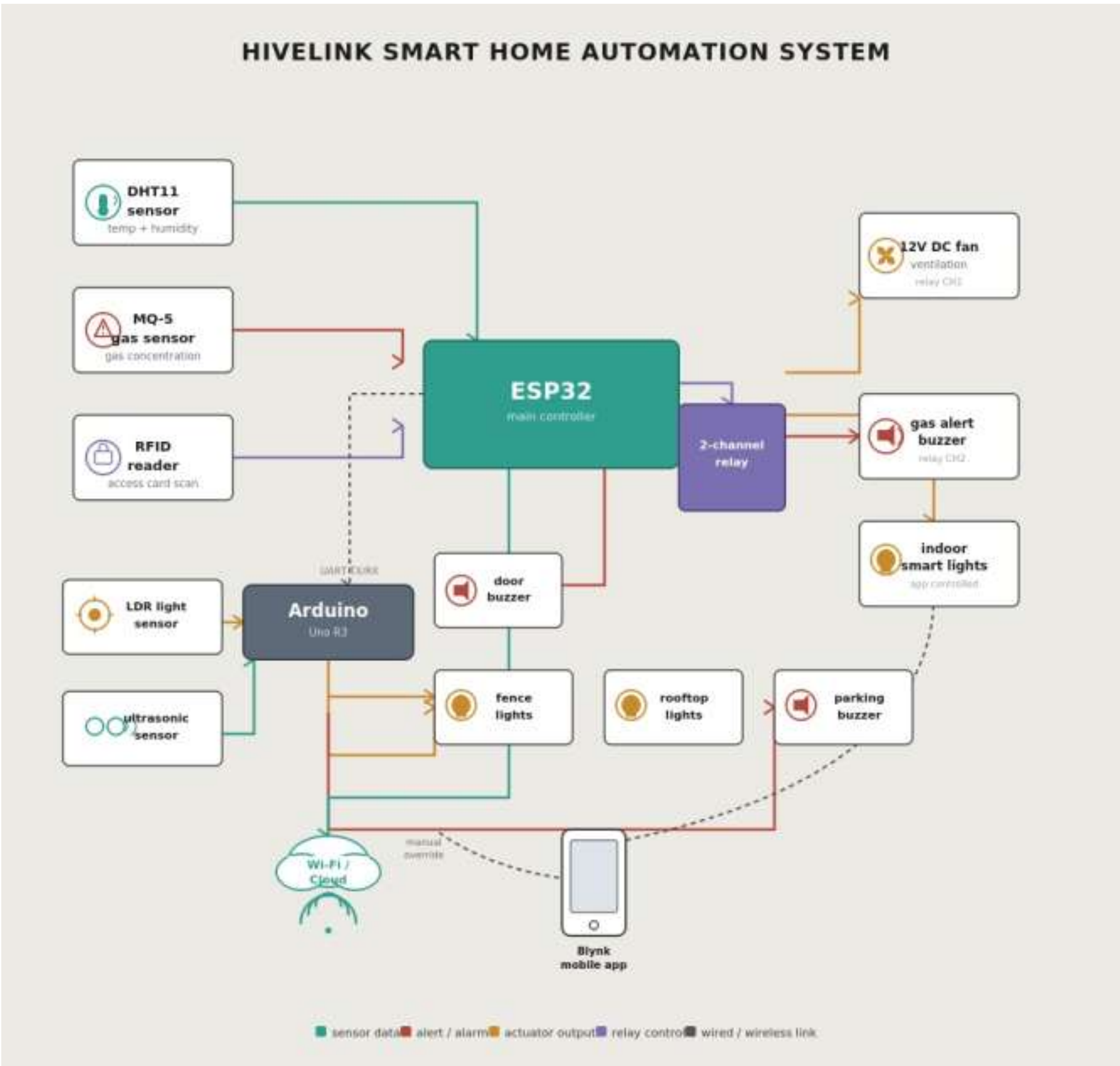
[ULT]

Smart Parking Assistance

Ultrasonic sensor measures the vehicle's distance from the garage wall. Activates a buzzer when distance drops to 2.9 cm or less, alerting the driver to stop safely.

System Architecture

The HiveLink system is powered by a dual-microcontroller architecture: an ESP32 handling all Wi-Fi, RFID, gas sensing, and kitchen monitoring functions, and an Arduino Uno managing outdoor lighting and parking assistance.



Architecture Overview

Layer	ESP32 Module	Arduino Uno Module
Sensing	DHT11, MQ-5, RFID Reader	LDR, Ultrasonic Sensor
Actuation	Relay + 12V Fan, Buzzer, Door Lock	Outdoor Lights, Buzzer
Display	LCD (16x2), Blynk App	—
Connectivity	Wi-Fi (Blynk IoT Cloud)	Serial / Local

Hardware Components

The following table lists all hardware components used in the HiveLink system:

#	Component	Specification / Role	Module
1	**ESP32 Microcontroller**	Main IoT brain with Wi-Fi, RFID & sensor control	ESP32
2	**Arduino Uno**	Outdoor lighting & parking management	Arduino Uno
3	**DHT11 Temperature & Humidity Sensor**	Measures kitchen temp (threshold 30 deg C) & humidity (threshold 60%)	ESP32
4	**MQ-5 Gas Sensor**	Detects LPG/combustible gas (threshold: 2000)	ESP32
5	**RFID Reader Module**	Reads authorized RFID cards/tags for door access	ESP32
6	**Relay Module**	Controls 12V DC ventilation fan	ESP32
7	**12V DC Fan**	Ventilation — activated by relay on high temp/humidity	ESP32
8	**LCD Display (16x2)**	Real-time local display of all sensor readings	ESP32
9	**Buzzer**	Alarm for gas leaks and parking distance alerts	ESP32 & Arduino
10	**LDR Sensor**	Ambient light sensing for outdoor lighting automation	Arduino Uno
11	**Ultrasonic Sensor**	Measures vehicle-to-wall distance for parking assist	Arduino Uno

Operating Instructions

Starting the System

1. Connect the ESP32 and Arduino Uno to their respective power sources.
2. Ensure the Wi-Fi credentials in the ESP32 firmware are correctly configured for your network.
3. Power on the system — the LCD will initialize and display live sensor readings within seconds.
4. Launch the Blynk app on your smartphone to begin remote monitoring.

Using RFID Door Access

1. Hold an authorized RFID card or tag near the RFID reader module.
2. If the credential is valid, the door lock relay activates and the door unlocks.
3. If an unauthorized card is presented, the system rejects it and the door remains locked.
4. Always register new RFID cards in the system firmware before use.

Monitoring the LCD Display

The 16x2 LCD provides a real-time local dashboard. It continuously shows:

- Temperature (in degrees Celsius)
- Humidity (in percentage)
- Gas Value (raw sensor reading)
- Gas Status - SAFE (normal) or ALERT (danger threshold exceeded)
- Fan Status - ON (ventilation active) or OFF (inactive)

Using the Blynk IoT Mobile Application

1. Open the Blynk app and log in to your account.
2. Select the HiveLink device from your dashboard.
3. View live temperature, humidity, and gas readings on your screen.
4. Monitor the gas status indicator - GREEN for SAFE, RED for ALERT.
5. Use the Light Control toggle to turn outdoor lights ON or OFF remotely.

Smart Outdoor Lighting (Automatic)

No manual action required. The LDR sensor continuously monitors ambient light levels:

- When light intensity drops below 100 (dusk/nighttime): Fence and rooftop lights turn ON automatically.
- When light intensity rises above 100 (daytime): Lights turn OFF automatically.

Smart Parking Assistance

When pulling a vehicle into the garage:

- The ultrasonic sensor begins measuring distance to the garage wall as soon as the vehicle enters range.
- As the vehicle approaches 2.9 cm from the wall, the buzzer activates as a stop signal.
- When the buzzer sounds, stop the vehicle immediately to avoid collision.

Mobile App Guide - Blynk IoT

The Blynk IoT platform serves as HiveLink's remote control and monitoring interface. Follow the steps below to get connected.

Step 1: Connecting to Wi-Fi

- Ensure your smartphone and the ESP32 are connected to the same Wi-Fi network during initial setup.
- Open the Blynk app and navigate to your device settings.
- Confirm that the device status shows Online before proceeding.

Step 2: Viewing Sensor Values

On the Blynk dashboard, the following values are displayed in real time:

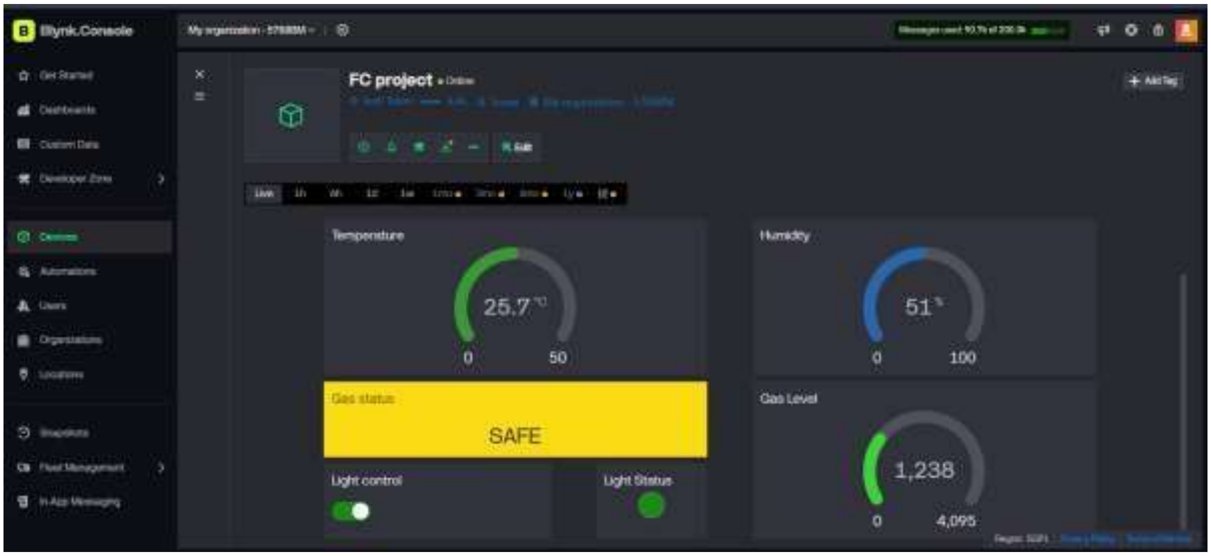
Widget	Data Shown	Update Frequency
Temperature Gauge	Current kitchen temperature (deg C)	Real-time
Humidity Gauge	Current kitchen humidity (%)	Real-time
Gas Value Display	Raw MQ-5 sensor reading	Real-time
Gas Status Label	SAFE (green) / ALERT (red)	Instant on change
Light Control Button	Toggle outdoor lights ON / OFF	On user action

Step 3: Remote Light Control

- Locate the Light Control toggle widget on the Blynk dashboard.
- Tap the toggle to turn outdoor lights ON or OFF from anywhere with an internet connection.
- The system responds within seconds of receiving the command.

Step 4: Understanding SAFE / ALERT Notifications

SAFE STATUS	ALERT STATUS
Gas value is below 2000. No hazard detected. System is operating normally.	Gas value is 2000 or above. Danger detected! Evacuate and ventilate the area immediately.


Blynk.Console
FC project
Temperature
25.7 °C
0
50
Humidity
51 %
0
100
Gas status
SAFE
Gas Level
1,238
0
4,095
Light control
Light Status

Product Advantages

HiveLink is built to deliver measurable value across safety, efficiency, and convenience:

Smart Monitoring 24/7 automated tracking of temperature, humidity, gas, light, and distance — no manual checks required.	**Enhanced Home Security** RFID door access eliminates key-based vulnerabilities and prevents unauthorized entry.
Energy Saving Light automation ensures electricity is used only when truly needed, reducing wastage.	**Remote Access** Full system visibility and control from any location via the Blynk IoT mobile application.
Real-Time Alerts Instant gas hazard notifications on LCD and mobile app — reacting faster than any manual system.	**User Friendly** Simple LCD interface, intuitive app controls, and automated responses require minimal user training.

Real-Life Applications

HiveLink is versatile and scalable - here is where it delivers the most value:

Icon	Application	Key Benefit
[HOME]	**Residential Homes**	Complete automation of lighting, security, kitchen monitoring, and gas safety.
[APRT]	**Apartments**	Compact and efficient — ideal for multi-unit buildings with shared garage spaces.
[KITCH]	**Smart Kitchens**	Real-time thermal and gas monitoring ensures safe and comfortable cooking environments.
[OFFC]	**Small Offices**	Automate entrance security and energy-saving lighting for small professional spaces.
[GARG]	**Garages & Driveways**	Parking assistance prevents vehicle collisions; motion-triggered outdoor lighting improves access.

Safety Precautions

IMPORTANT - Please Read Before Use

Failure to follow these safety guidelines may result in system malfunction, property damage, or personal injury.

Electrical Safety

- Ensure all components are powered with the correct voltage (5V for ESP32/Arduino; 12V DC for the fan).
- Never expose the circuit boards or wiring to water or excessive moisture.
- Always power OFF the system before modifying connections or replacing components.
- Use a surge protector to protect against voltage spikes.

Gas Safety

- If a GAS ALERT is triggered, immediately turn off all gas appliances and open windows and doors for ventilation.
- Do NOT use open flames, switches, or electronic devices in the area until the gas alert clears.
- Contact emergency services if the ALERT persists after ventilation.
- Test the MQ-5 sensor periodically to ensure it is functioning correctly.

RFID Security

- Keep RFID cards in a secure location. Report lost or stolen cards immediately and deauthorize them from the system.
- Do not share RFID credentials with unauthorized individuals.

General

- Keep all sensors clean and free from dust or obstructions for accurate readings.
- Do not tamper with the hardware components unless you are a qualified technician.
- Place the MQ-5 gas sensor in a location with good air circulation for reliable detection.

Troubleshooting Guide

If you experience any issues with HiveLink, refer to the table below for common problems and solutions:

Problem	Possible Cause	Solution
LCD shows no data	Power supply issue or loose wire	Check power connections and verify ESP32 is properly powered.
Gas ALERT triggered with no gas leak	Sensor exposed to dust or moisture	Clean the MQ-5 sensor and ensure it is placed in a ventilated area.
Blynk app shows device as Offline	Wi-Fi credentials incorrect or network down	Verify Wi-Fi SSID and password in firmware; check your internet connection.
RFID card not recognized	Card not registered in firmware	Add the RFID card UID to the authorized list in the ESP32 sketch.
Ventilation fan does not activate	Relay module connection faulty	Inspect relay wiring and confirm thresholds are set correctly in code.
Outdoor lights stay ON during daytime	LDR sensor blocked or damaged	Check LDR placement for obstructions and test sensor with a multimeter.
Parking buzzer activates too early	Ultrasonic sensor misaligned	Re-angle the sensor to point directly at the garage wall; recalibrate threshold.
System restarts randomly	Insufficient power supply	Use a stable 5V/2A power adapter; check for loose USB connections.

Future Enhancements

HiveLink is designed as a scalable platform. The following enhancements are planned for future versions:

[VOC]

Voice Assistant Integration

Integrate with Amazon Alexa, Google Assistant, or Apple Siri for hands-free home control — control lights, check gas status, and lock doors with voice commands.

[AI]

AI-Based Predictive Automation

Use machine learning to learn household patterns and proactively adjust ventilation, lighting, and security settings before the user needs to act.

[PUSH]

Mobile Push Notifications

Instant push alerts directly to your smartphone for gas leaks, unauthorized RFID attempts, or temperature anomalies - no app open required.

[CLOUD]

Cloud Data Analytics Dashboard

Historical sensor data stored in the cloud with trend graphs, usage analytics, and energy consumption reports available on a web dashboard.

[CAM]

Camera & Video Integration

Add IP cameras to monitor doors, garage, and kitchen remotely- with motion detection alerts sent directly to the Blynk app.

[SOLAR]

Solar & Renewable Power Support

Power outdoor lighting and sensors from solar panels - making HiveLink a zero-carbon smart home solution.

Conclusion

HiveLink is not just a university project.

It is a scalable, real-world smart home solution designed to create safer, smarter, and more connected living spaces - today and into the future.

HiveLink brings together the power of IoT, embedded systems, and cloud connectivity to deliver a product that makes a meaningful difference in everyday life. Every feature has been thoughtfully engineered - from the gas sensor that guards your kitchen, to the RFID lock that secures your front door, to the parking assistant that protects your vehicle.

What began as an academic project represents something far greater: a proof of concept that students and developers can build technology that competes with commercial smart home products. HiveLink is modular, expandable, and ready to grow with the needs of modern households.

As smart home technology continues to evolve, HiveLink is positioned to integrate voice assistants, AI-driven automation, renewable energy, and advanced analytics - transforming from an intelligent home companion to a fully autonomous living ecosystem.

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